



OneATS

User Manual

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OneATS user manual

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Definitions, symbols, abbreviations

Symbol	Description
CIM	Common Information Model
DMS	Distributed Management System
DCOM	Distributed Component Object Model
FEP	Front End Processing
HMI	Human Machine Interface
I/O	Input/Output
IED	Intelligent Electronic Device
LAN	Local Area Network
OPC UA	OPC Unified Architecture
RTU	Remote Terminal Unit
SCADA	Supervisory Control and Data Acquisition
SSL	Secure Socker Layer
TCP	Transmission Control Protocol
TLS	Transport Layer Security
XML	Extensible Markup Language

Text color:

Comment

Notes

Warnings

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A.	System overview
B.	Database
C.	OneATS DataEditor – Project Management
D.	OneATS FEP
E.	OneATS DataServer
F.	OneATS Grid Studio
G.	OneATS SmartHIS
H.	OneMasterTool



OneATS User Manual

- OneATS DataEditor – Project Management

14/Jan/2025

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Software Version	Date of Issue	Change Description
V4.2.3.8	30/May/2024	
V4.2.3.51	31/Dec/2024	-Implement PSM (Protection Settings Manager) setting configuration - Implement new database copy method for mongodb version >= 4.2 - Not allow add Region with structured

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C. OneATS DataEditor – Project Management

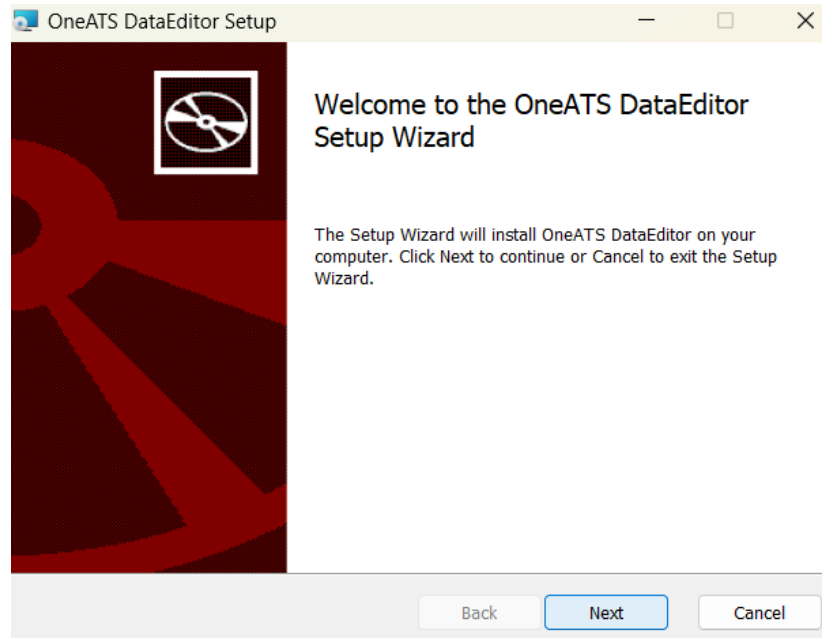
C.1. OneATS DataEditor Overview

OneATS DataEditor is a module in the OneATS product suite, which is a tool for building and configuring system data, including:

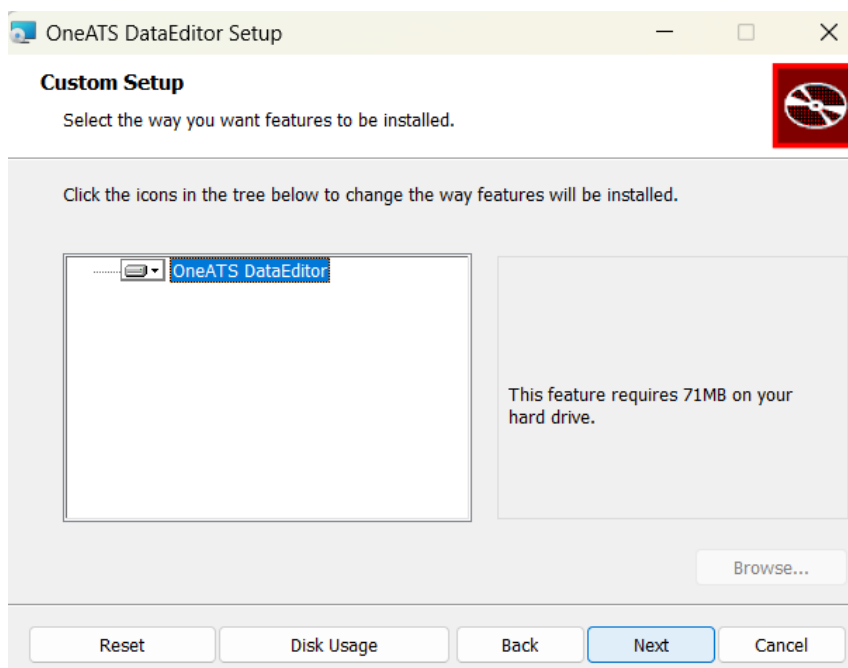
- Build FEP data channels (IEC61850, IEC101, IEC103, IEC104, Modbus, OPC, Sel Fast Message, ...).
- Build a SCADA/DMS DataModel.
- System configuration: declare and connect computers performing functions (HMI, DataServer, HIS, FEP, ...) according to redundancy and replication requirements.
- Configure Data Mapping.
- Realtime FEP data tracking.
- DataServer realtime data tracking.

C.2. Install the app

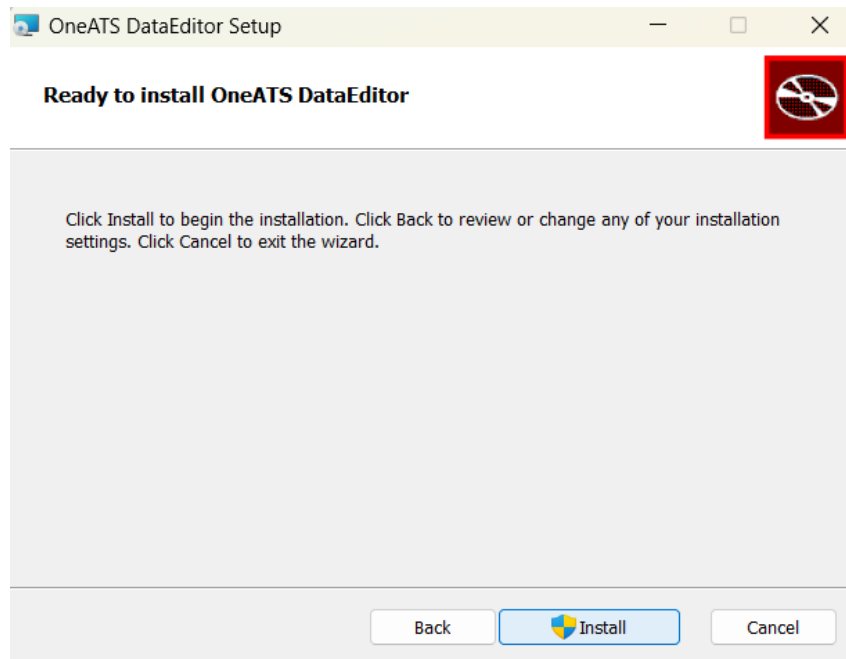
Run the ATS OneATS installation file DataEditor.msi. Select **Next** to continue:



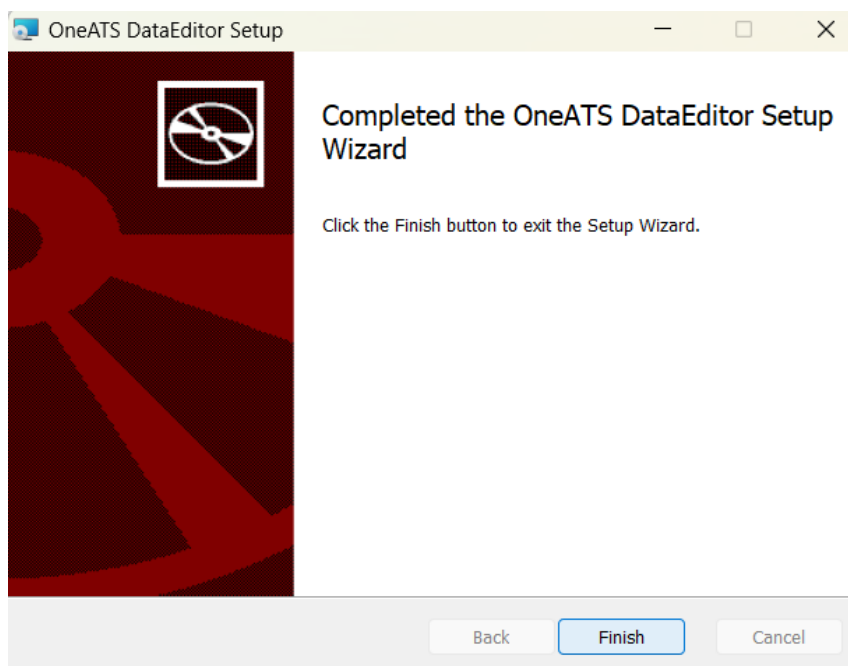
Select **Next** to continue:



Click **Install** and wait for the installation process to take place:



Select **Finish** to finish the installation process:



C.3. Project Management with OneATS DataEditor

C.3.1. OneATS DataEditor Interface

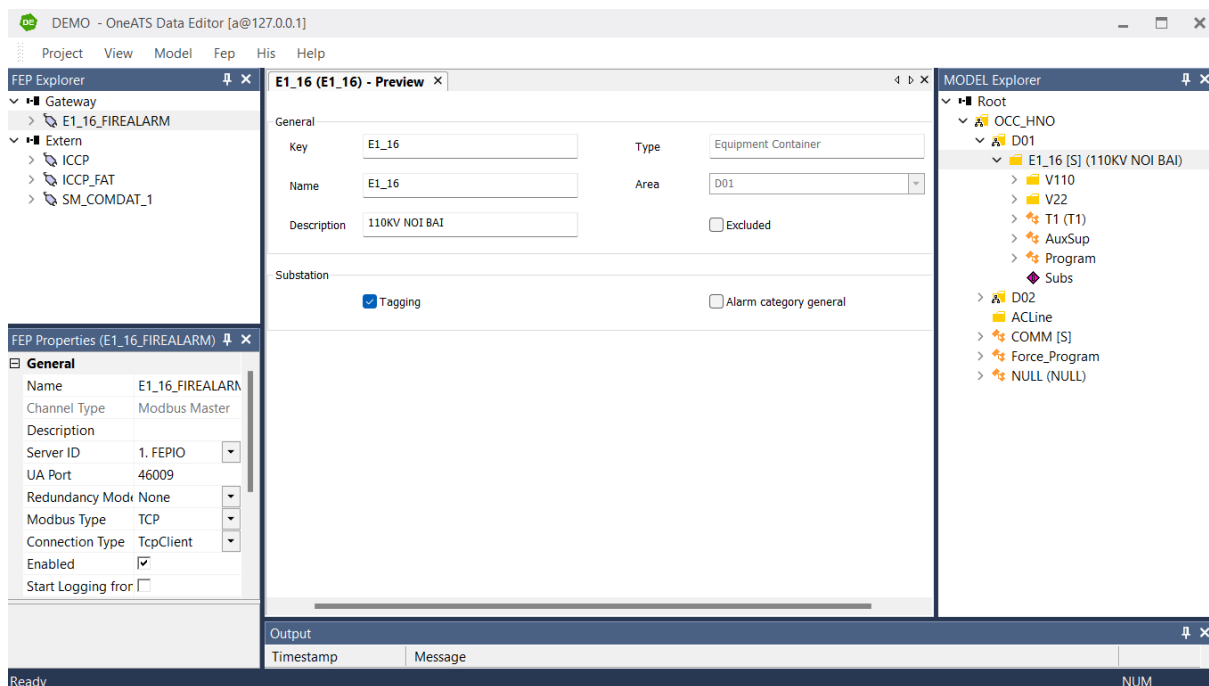
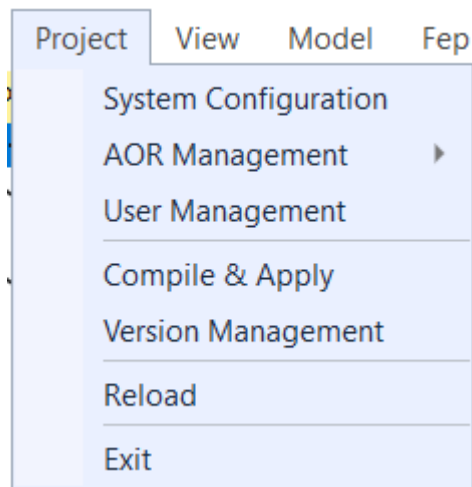


Figure C-1. DataEditor Main Screen

The basic interface of the DataEditor consists of 3 main areas:

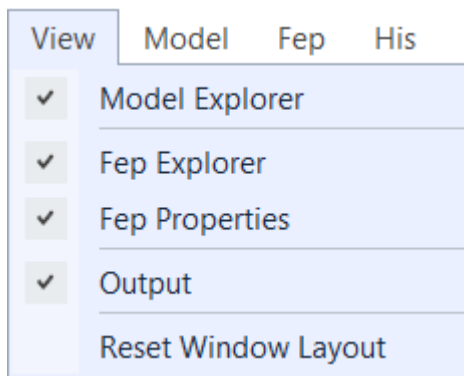
- **FEP Explorer:** is responsible for managing *FEP data channels*.
- **MODEL Explorer:** is responsible for managing the data model (DataModel).
- **Data configuration area:** located between FEP Explorer and MODEL Explorer is responsible for performing calculations, mapping, data transformation, etc. between *FEP* and *DataModel channels* in both directions.
- **Menu Bar**

C.3.1.1 Menu Bar – Project



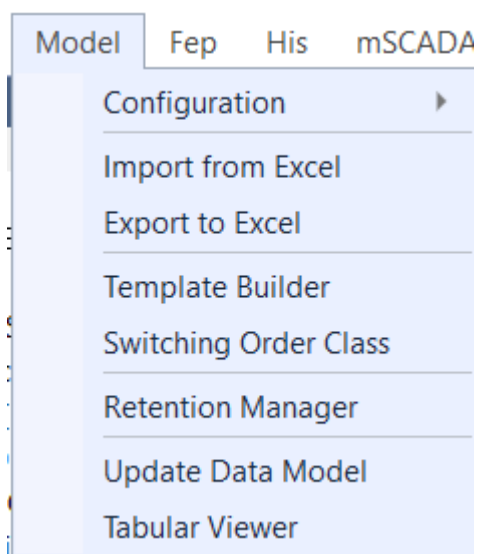
- Please refer Item C.4 to configure project
- Please refer Item C.5 to go to AOR Management
- The User Management is available in HMI module only
- Refer Item C.3.6 to compile & Apply project
- Version Management: Manage version of project
- Reload: Reload project
- Exit: Exit project

C.3.1.2 Menu Bar – View



Check/Uncheck to reset the layout.

C.3.1.3 Menu Bar – Model



- Configuration Model. Please refer in Document - Chapter E.
- Import from Excel: Import Excel Model file
- Export to Excel: Export Model to excel file
- Template Builder: Configure Template. Please refer in—Chapter E
- Switching Order Class: Refer to Chapter E
- Retention Manage: Manage all Retentive model tag
- Update Data Model: Update Model to DataServer – Item C.3.7

C.3.1.4 Menu Bar – FEP

Refer to Chapter D – OneATS FEP

C.3.1.5 Menu Bar – HIS

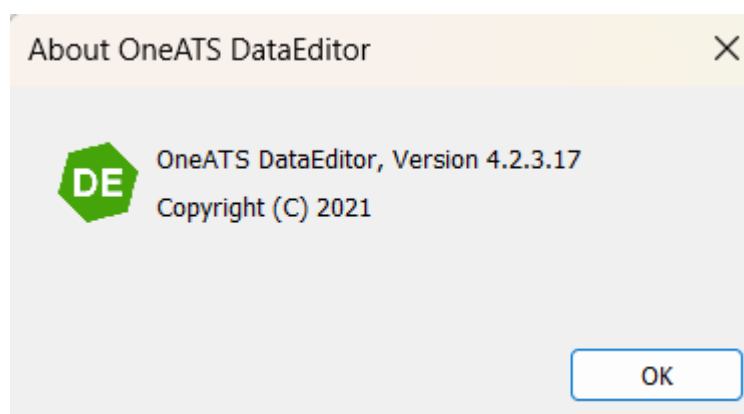
Refer to Chapter G – OneATS HIS

C.3.1.6 Menu Bar – mSCADA

Growing features

C.3.1.7 Menu Bar – Help

Help >> About >>



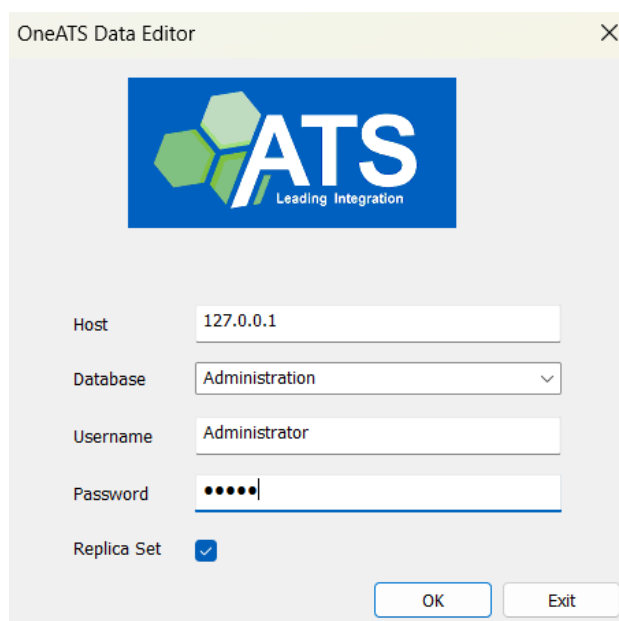
C.3.2. Log in to the database

Run **OneATS DataEditor** → to enter the parameters.

Parameter	Meaning	Example
Database	IP address, TCP port of (MongoDB) Database. Where there is no TCP port parameter, the application will use the default port 27017 .	127.0.0.1 (TCP Default Port 27017)
Username	UserName, Password Administrator List of Projects (MongoDB), or a Project (Collection)	
Password		
Login to	Administration: Open the project management window [Project Name] opens the project configuration window	
Replica Set	MongoDB can deploy replication to increase redundancy, the database will then be stored on multiple machines. MongoDB recommends a replication configuration with an odd number of machines (e.g. 3 machines, 5 machines, 7 machines, etc.). If the number of machines left in the system is less than 1/2 of the total number of machines, the system is only in Readonly mode, and cannot be edited. The DataEditor can only edit data on the Primary machine, while if it is connected to the Secondary machine, it will only view the data. "ReplicaSet", if enabled, the program will automatically find the Primary machine in the replication system to connect to.	

Default User: Administrator

Default Password: atsc0



The screenshot shows the 'OneATS Data Editor' login window. It features the OneATS logo at the top, which includes the text 'Leading Integration'. Below the logo are input fields for 'Host' (127.0.0.1), 'Database' (Administration), 'Username' (Administrator), and 'Password' (masked with dots). There is a 'Replica Set' checkbox which is checked. At the bottom right are 'OK' and 'Exit' buttons.

Figure C-2. Database login window

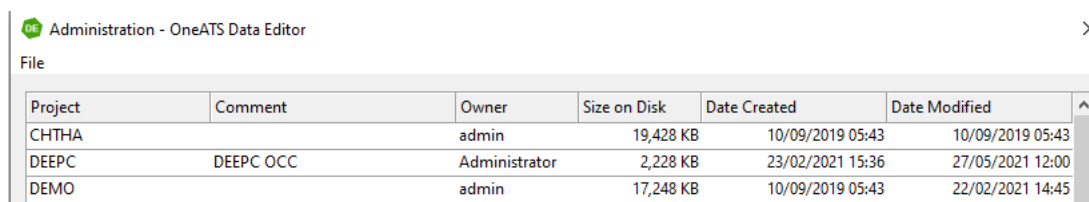
C.3.3. Project Management

Run the OneATS DataEditor app.

Log in with the following information:

Host	The (IP:Port) address of the (MongoDB) database
Database	Administration
Username/Password	user/password log in to MongoDB database
Replica Set	Select Tick

Click **OK** to open the project management window.



The screenshot shows the 'Administration - OneATS Data Editor' window. It has a 'File' menu and a table with project data.

Project	Comment	Owner	Size on Disk	Date Created	Date Modified
CHTHA		admin	19,428 KB	10/09/2019 05:43	10/09/2019 05:43
DEEPC	DEEPC OCC	Administrator	2,228 KB	23/02/2021 15:36	27/05/2021 12:00
DEMO		admin	17,248 KB	10/09/2019 05:43	22/02/2021 14:45

Figure C-3. Project Management Window

C.3.4. Create a new project

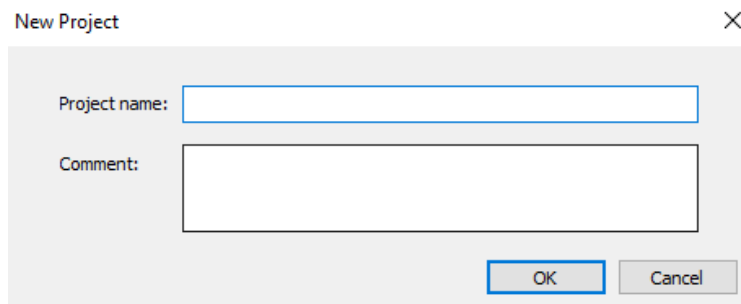
Open the project management window.

Click File and → select **New project**.

Enter a name for the project in the **Project name** section.

Add notes and descriptions in the **Comment** section.

Click **OK**. The newly created project will appear in the project management window.



A dialog box titled "New Project" with a close button (X) in the top right corner. It contains two input fields: "Project name:" and "Comment:". Below the input fields are two buttons: "OK" and "Cancel".

C.3.5. Open a project

To open a project, do the following:

Open the database login window.

In the Database section, fill in the name of the project you want to open.

Click OK .

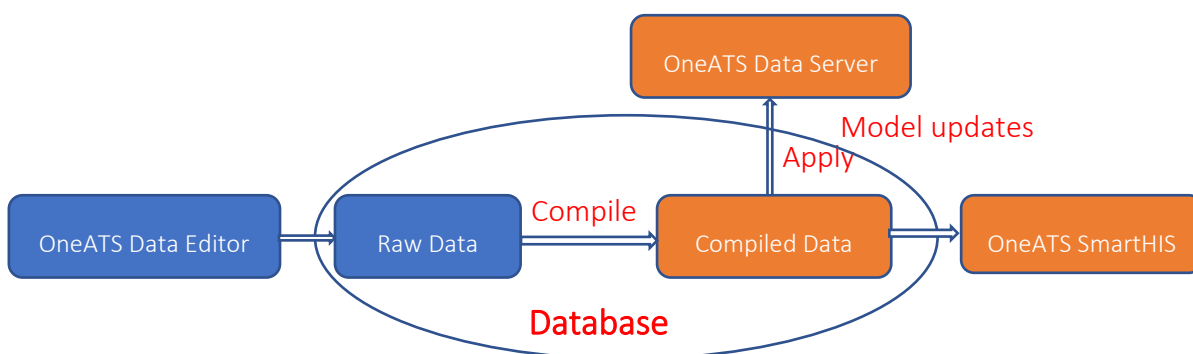
Or:

Open the project management window.

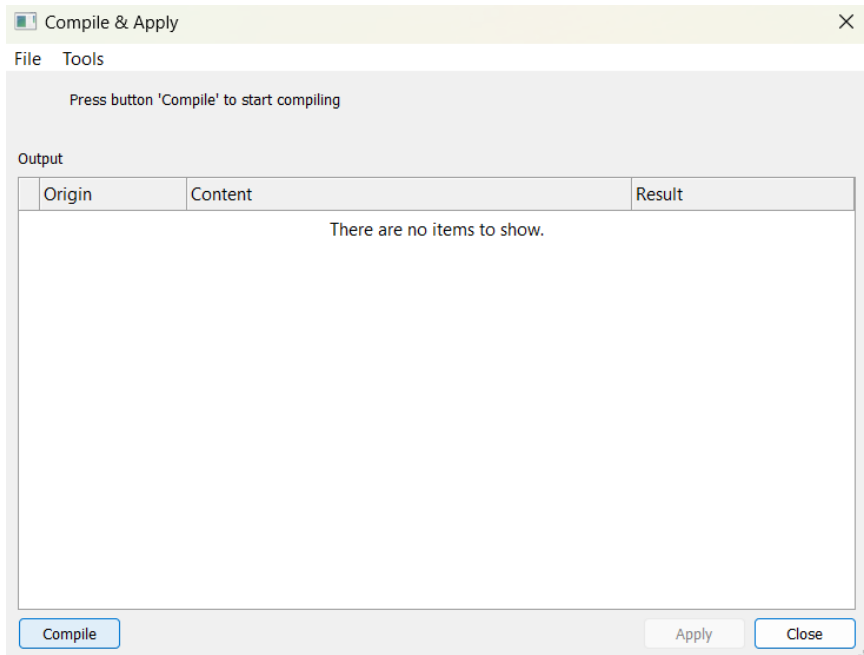
Double Click on the project you want to open.

C.3.6. Compile project

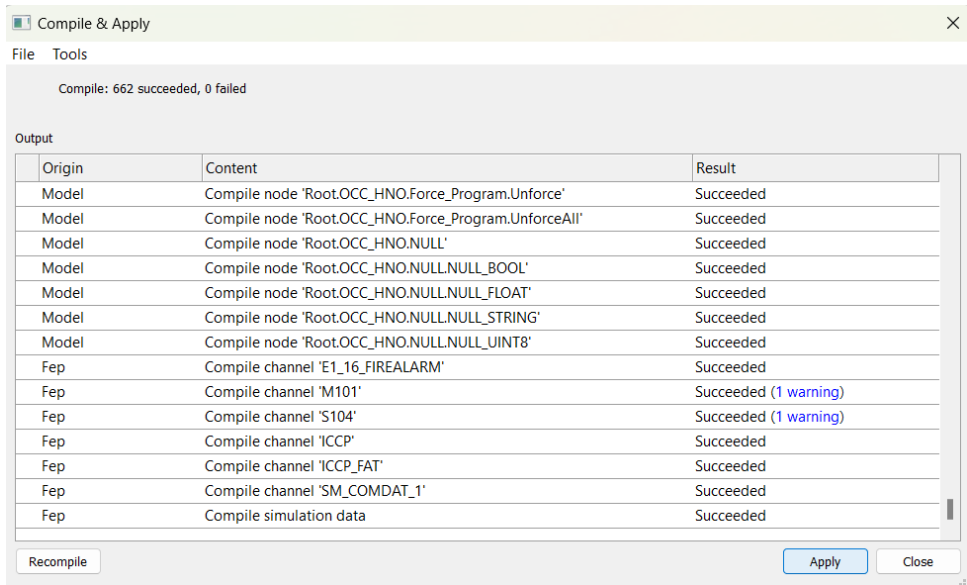
The Compile/Apply process is illustrated as shown below:



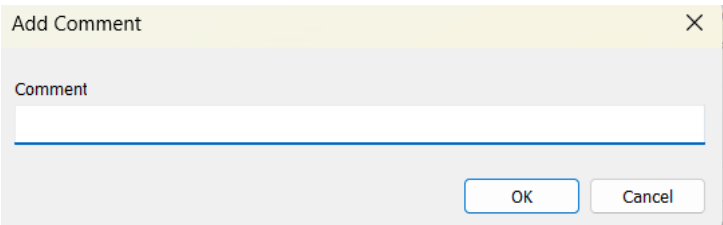
On the DE toolbar, select **Project/Compile & Apply**.



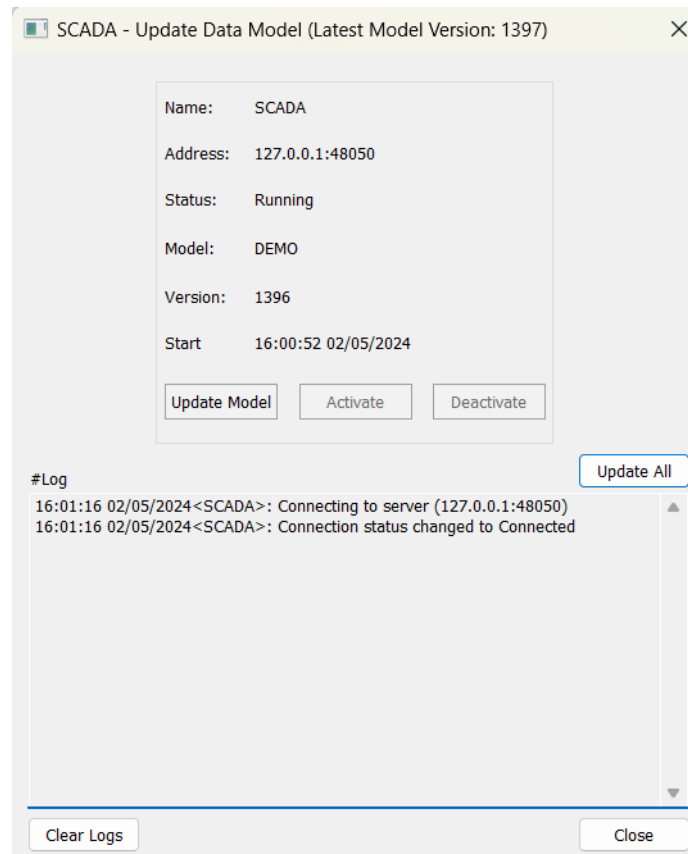
Select **Compile**, and if no errors are detected, select **Apply**.



Enter **a comment**, select **OK**.

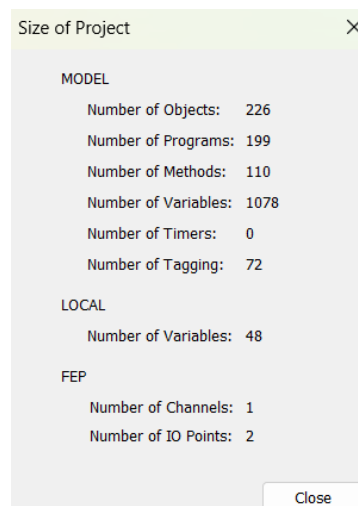


Choose **Model Updates** Servers one by one or select **Update All** for all servers. Details in the section C.3.7.



In addition, users can perform the following operations:

- File /**Size of Project Tab**: Statistics of Project information



- File /**Change Logs tab**: Statistics of changes of the current data model compared to the old data model

Change Logs		
Category	Origin	Change Type
Alarm Group	Voltage	Delete
Alarm Group	Current	Delete
Alarm Group	Reactive Power	Delete
Alarm Group	Active Power	Delete
Alarm Group	Current	Add
Alarm Group	Voltage	Add
Alarm Group	Active Power	Add
Alarm Group	Reactive Power	Add
Template	PTC	Update
Template	GE_L90	Update
Template	SIEMENS_7SL62	Update

- File /Exit Tab: Close the Compile & Apply window
- Tools /Options tab: Compile configuration information

Compiler Configuration

Compiler version:

Latest (V4.0)

☐ Compile automatically when opening
 ☐ Bypass name validation (not recommended)

Script compilation warning level

- Syntax error & exception:

Warning

- Invalid trigger item:

Warning

IEC61850 refined model level:

Data Object

Output:

All

☒ Integrate alarm limit object
 ☒ Refine data model (remove unused nodes)

Template configuration warning level

- Missing input tag:

Warning

- Invalid input tag:

Warning

- Missing input text:

Warning

- Missing input constant:

Warning

- Missing input sound:

Warning

OK

Cancel

Figure C-4. Compiler configuration

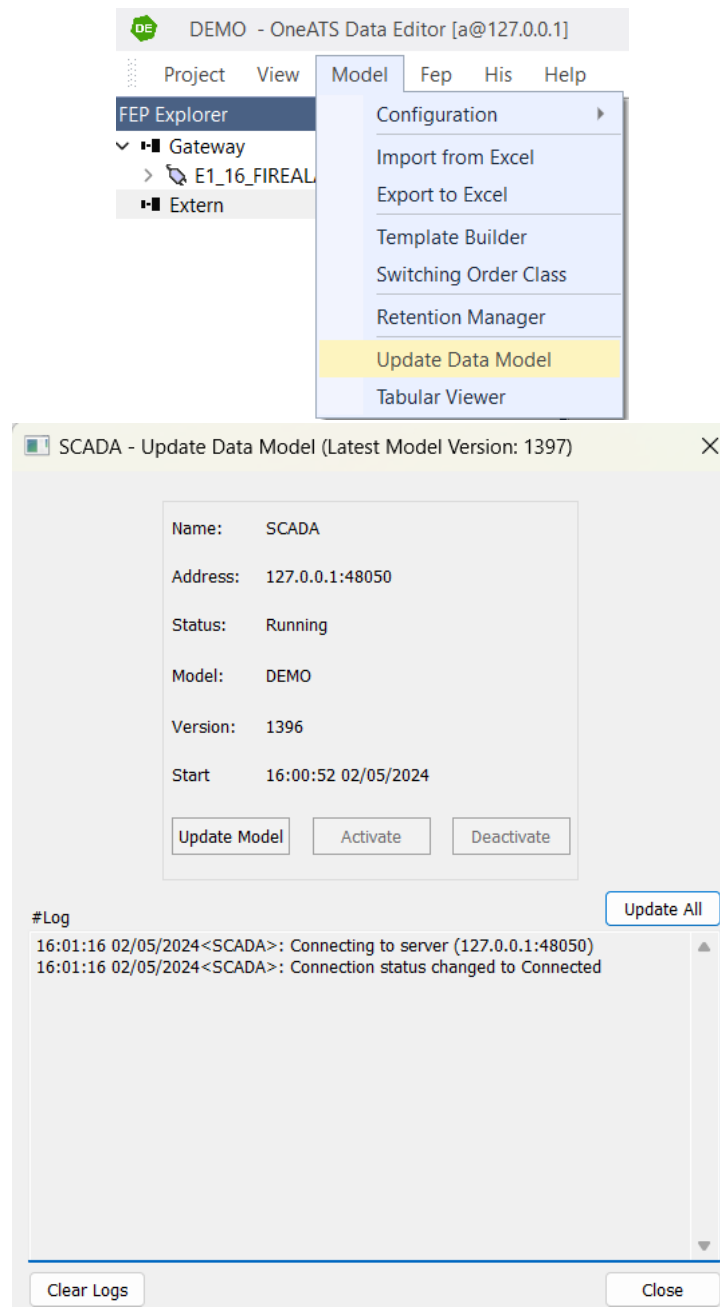
- Compile automatically when opening project
- Bypass validation
- Script compilation warning level:
 - Syntax: Warning/Disable/Error
 - Invalid trigger item: Warning/Disable/Error
- Integrate alarm limit object
- Refine data model:
- Template configuration warning level: Warning/Disable/Error

C.3.7. Update Data Model

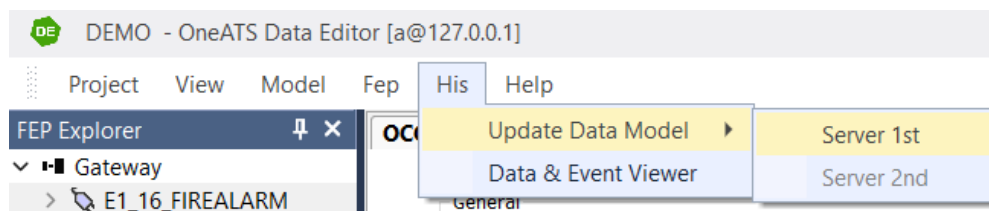
The Update Data Model feature allows the system to update data model changes to the DataServer without having to stop and rerun the service. The steps to perform the update are as follows:

After calibrating the Data Model on DE → Compile and Apply project.

Update Dataserver: On the DE toolbar, select **Model/Update Data Model**, select **Update Model** for Servers in turn, or select **Update All** for all servers.



Update **Data Model** for Hisservice: On the DE toolbar, select **HIS/Update Data Model**, select the Servers in turn.

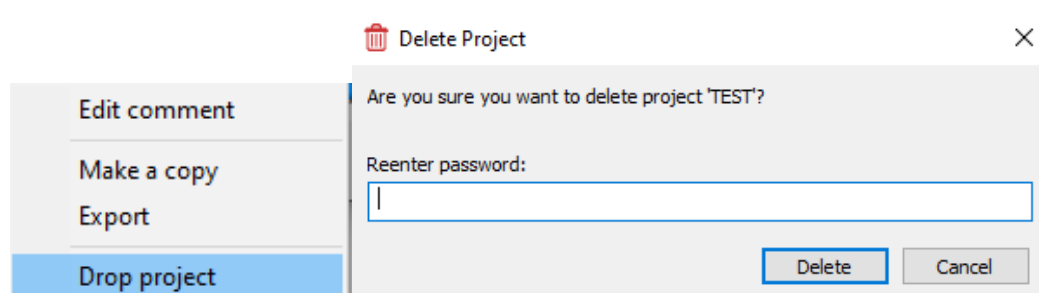


C.3.8. Delete a project

Open the project management window.

Click Right Click on the project you want to delete, select **Drop project**.

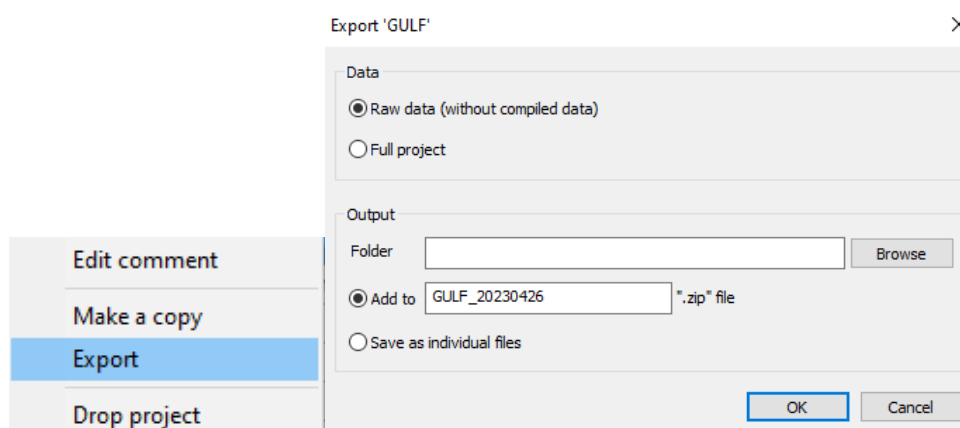
Enter your login password, then select **Delete**.



C.3.9. Export project

Open the project management window.

Click Right Click on the project you want to export, select **Export**.



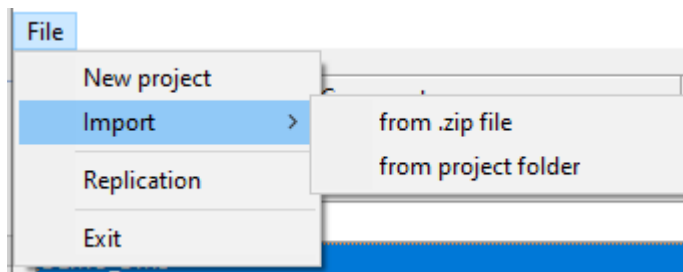
In which:

- **Raw data (without compiled data):** Includes only the portion of data configured by the user, not the data compiled by the software.
- **Full project:** All the data of the project, including the data that the software compiles.
- **Folder:** Choose a location to save the export file.
- **Add to:** Save the export file as a .zip file.
- **Save as individual files:** Save the export file as a folder.

C.3.10. Import project

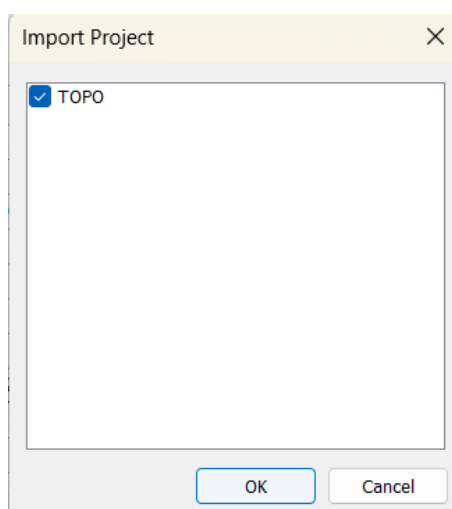
Open the project management window.

Select **File**, select **Import**, select import **from.zip file** or **from project folder**.



Select the file or folder where the project's data is saved.

Check the project you want to import, select **Ok**.



After successfully importing the project, it will appear in the project management window.

C.3.11. Backup project

Operate like Export project.

C.3.12. Revise project descriptions

Open the project management window.

Click Right Click on the project you want to edit the description, select **Edit comment**.

Enter a new description name, get **Ok** to make the correction.



Edit comment

Make a copy

Export

Drop project

Edit Comment

Project name: GULF

Comment: GULF OCC

OKCancel

C.4. Project Configuration

Open the project, go to the Project/*System Configuration path* to configure the system, the configurations include:

- SCADA System.
- FEP System.
- HIS System.

C.4.1. Concept of Redundancy

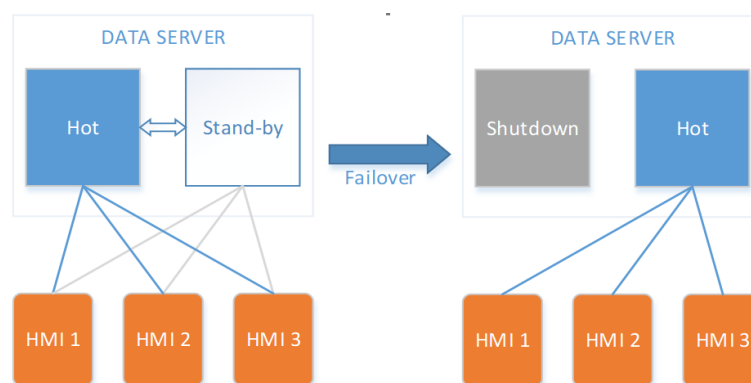


Figure C-5. Configuring Redundancy System

- (Server) Redundancy is a feature that configures the redundancy service of the system.
- Data is deployed at different computers.
- When the main service is digitalized, the system will automatically switch to using a backup service.
- The system ensures the synchronization and consistency of data when the computer crashes and recovers.
- The system fully supports Redundancy in Server-side modules: FEP, DataServer, HIS, DataCenter.

C.4.2. SCADA System Configuration

C.4.2.1 Add a new SCADA Server

In the **Project → System Configuration → SCADA System** window, → select **Add server**. Fill in the configuration parameters:

- **Server Name:** the name of the server.
- **Primary Server URL:** the server address and data port of the primary server.
- Check **Redundancy Server Deployment** if you want to deploy redundancy.
- **Redundancy Server URL:** the address and data port of the redundancy server.

- **Failover Time:** the time of failover.
- **Auto-Revert Enabled:** Determines what happens when the primary server comes back online after the failover takes place. When checked, the primary server will resume the running mode when it comes back online, the backup server will switch to the standby mode..

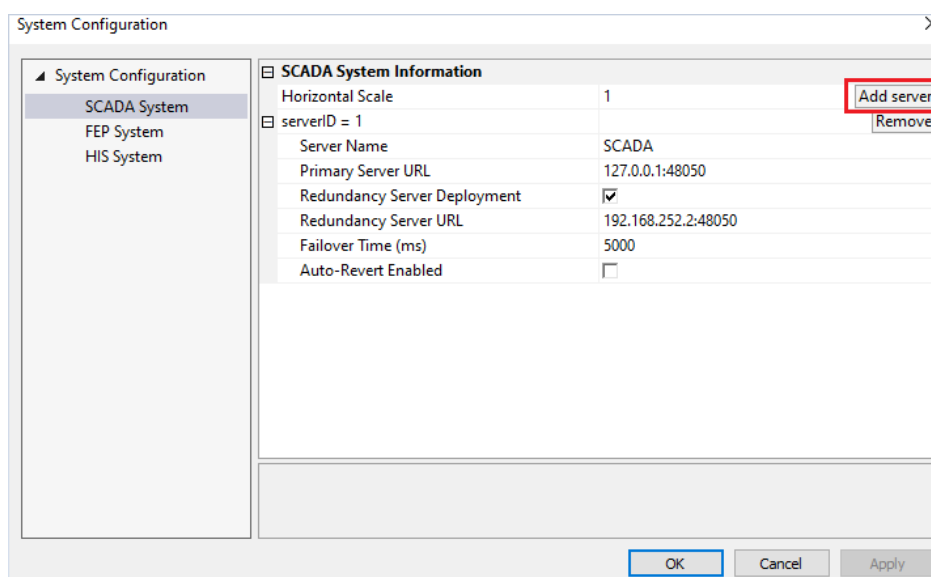


Figure C-6. SCADA System Configuration

C.4.2.2 Delete SCADA server

In the **Project** → **System Configuration** → **SCADA System** window, → select **Remove** next to the serverID line that you want to remove.

C.4.3. FEP System Configuration

The operations to add and delete the server are similar to the SCADA System configuration.

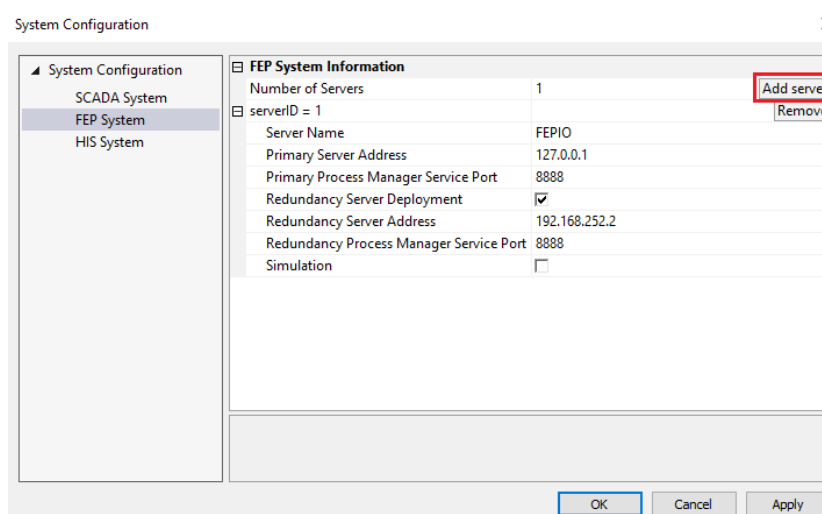


Figure C-7. FEP system configuration

- **Simulation:** Data simulation function, refer to ChapterE.OneATS DS.

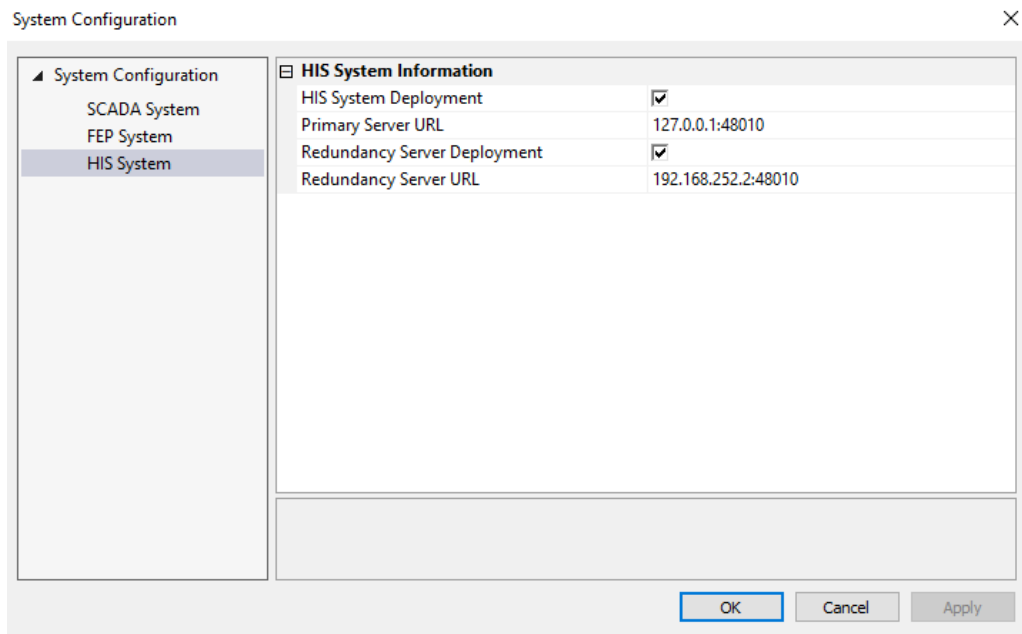
C.4.4. HIS System Configuration

In the **Project → System Configuration → HIS System →** window, select **HIS System Deployment** to deploy the HIS server.

Check **Redundancy Server Deployment** if you want to deploy Redundancy.

Fill in the parameters:

- **Primary Server URL:** the IP address and port of the primary server.
- **Redundancy Server URL:** The IP and port of the redundancy server.



The screenshot shows the 'System Configuration' dialog box with the 'HIS System Information' tab selected. The dialog has a left sidebar with 'System Configuration', 'SCADA System', 'FEP System', and 'HIS System'. The main area contains a table with the following data:

HIS System Information	
HIS System Deployment	☒
Primary Server URL	127.0.0.1:48010
Redundancy Server Deployment	☒
Redundancy Server URL	192.168.252.2:48010

At the bottom right, there are three buttons: 'OK', 'Cancel', and 'Apply'.

Figure C-8. HIS system configuration

C.5. AOR Management

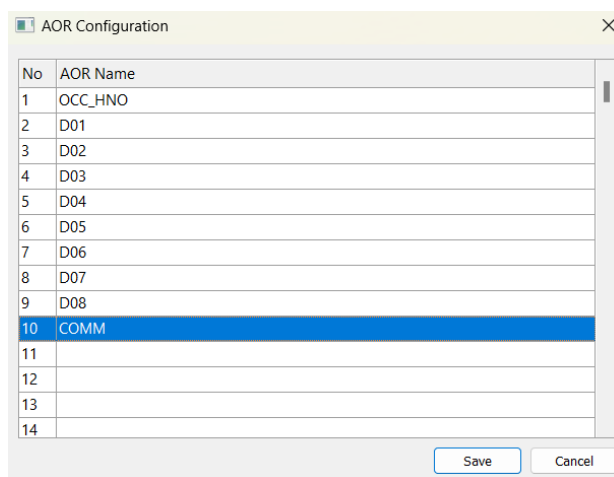
C.5.1. AOR (Area of Responsibility) Management

The AOR is the area of responsibility, each AOR contains Bays/Transformers and is fully responsible for their devices.

AOR Declaration Techniques in the Region of Any Object

Open the **Project → AOR Management window**, select **AOR**. The AOR Configuration **window** appears.

Double Click on the blank box of the **AOR Name column**, enter the name of the AOR Name, select **Save** to save the configuration.



No	AOR Name
1	OCC_HNO
2	D01
3	D02
4	D03
5	D04
6	D05
7	D06
8	D07
9	D08
10	COMM
11	
12	
13	
14	

Save Cancel

Figure C-9. AOR Management

C.5.2. AOR Group Management

Each AOR Group contains many different AORs, Groups help Users recognize their area of responsibility.

Open the **Project → AOR Management window**, select **AOR Group**. The AOR Group Configuration **window** appears.

Double Click on the blank box of the **AOR Group Name column**, enter the name of the AOR Group Name, add the AORs to the AOR Group by **turning ON** the AORs belonging to the AOR Group, select **Save** to save the configuration.

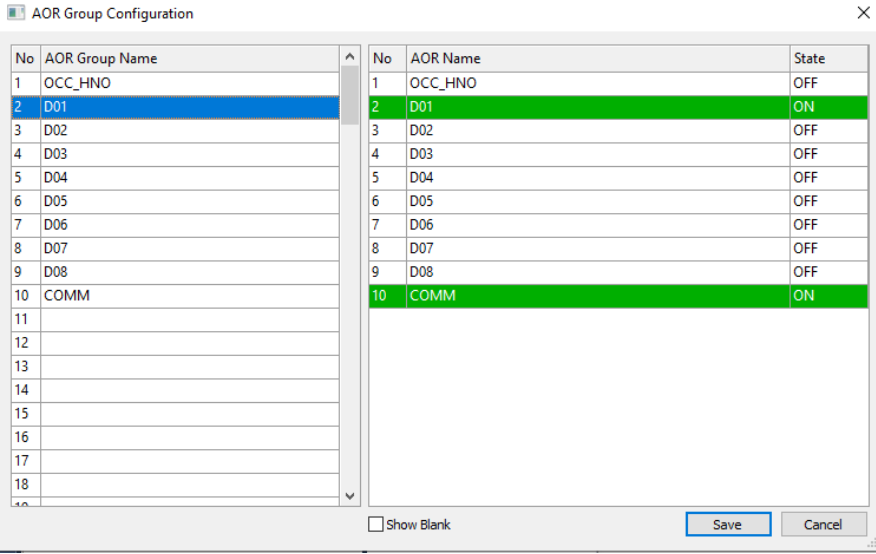


Figure C-10. AOR Group Management

C.5.3. Delete AOR/AOR Group

Open the **Project → AOR Management** window, select **AOR/AOR Group**.The AOR Configuration/AOR Group Configuration **window** appears, delete the names of the AOR/AOR Groups to be deleted, and select **Save** to save the configuration.

C.6. User Management

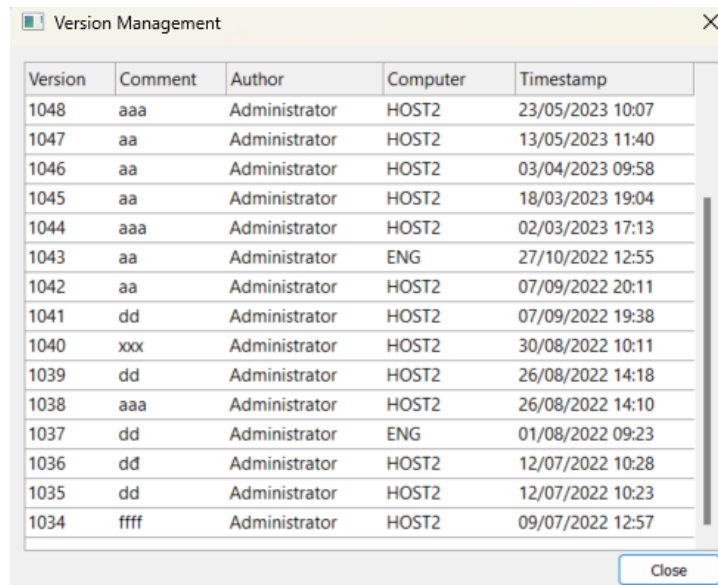
The function of managing users in the system, currently only configurable in OneATS Grid Studio.

Refer to Chapter H-HMI to use User Management.

C.7. Manage project versions

C.7.1. Manage project versions

Open the **Project → Version Management** window. The **Version Management** window appears.



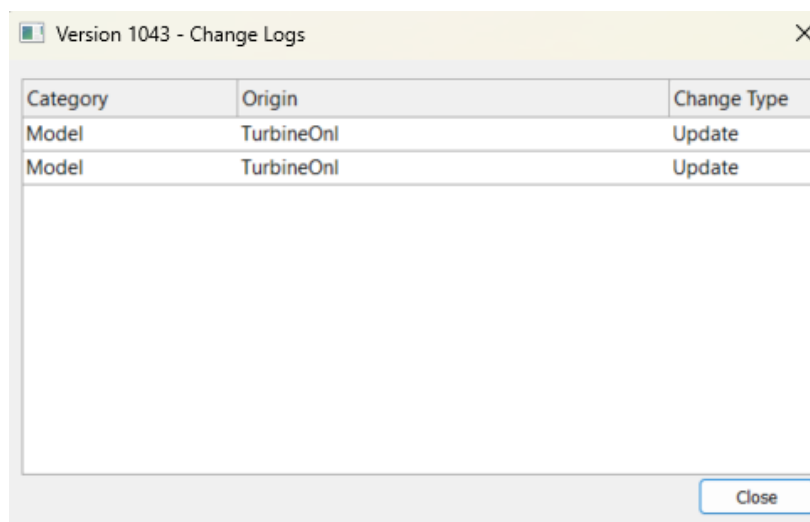
Version	Comment	Author	Computer	Timestamp
1048	aaa	Administrator	HOST2	23/05/2023 10:07
1047	aa	Administrator	HOST2	13/05/2023 11:40
1046	aa	Administrator	HOST2	03/04/2023 09:58
1045	aa	Administrator	HOST2	18/03/2023 19:04
1044	aaa	Administrator	HOST2	02/03/2023 17:13
1043	aa	Administrator	ENG	27/10/2022 12:55
1042	aa	Administrator	HOST2	07/09/2022 20:11
1041	dd	Administrator	HOST2	07/09/2022 19:38
1040	xxx	Administrator	HOST2	30/08/2022 10:11
1039	dd	Administrator	HOST2	26/08/2022 14:18
1038	aaa	Administrator	HOST2	26/08/2022 14:10
1037	dd	Administrator	ENG	01/08/2022 09:23
1036	dd	Administrator	HOST2	12/07/2022 10:28
1035	dd	Administrator	HOST2	12/07/2022 10:23
1034	ffff	Administrator	HOST2	09/07/2022 12:57

Figure C-11. Project version management

C.7.2. Manage changes in project versions

Open the **Project → Version Management** window. The **Version Management** window appears.

Click Right Click on the version you want to see the changes from the previous version, select **Show change logs**.



Category	Origin	Change Type
Model	TurbineOnl	Update
Model	TurbineOnl	Update

Figure C-12. Manage changes from previous versions of a project

C.7.3. Correcting comments in the project version

Open the **Project → Version Management** window. The **Version Management** window appears.

Click Right Click on the version you want to edit the comment for, select **Edit comment**.

Re-enter a new comment, select **Ok** to save it.

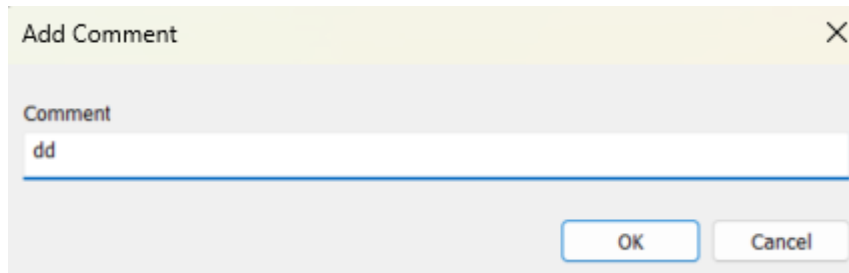


Figure C-13. Correcting comments in the project version

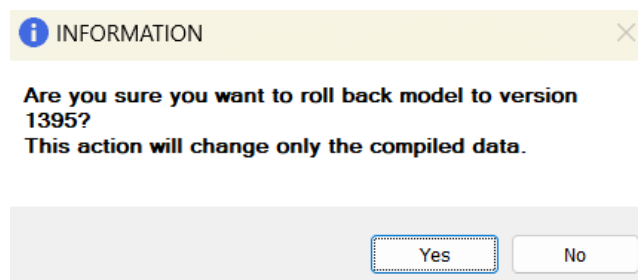
C.7.4. Roll back feature

The Roll back feature allows you to go back to previous versions of DataServer.

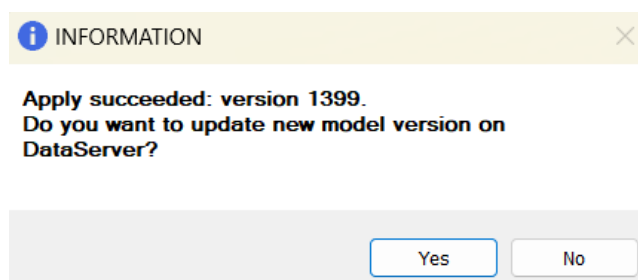
Open the **Project → Version Management** window. The **Version Management** window appears.

Click Right Click on the version you want to Roll back, select **Roll back**.

The **INFORMATION** window appears, select **Yes**.



Select **Yes** to update the new model to DataServer.



Select **Update Model** for servers in turn or select **Update All** for all servers.



SCADA - Update Data Model (Latest Model Version: 1399)

Name: SCADA

Address: 127.0.0.1:48050

Status: Running

Model: DEMO

Version: 1396

Start 16:00:52 02/05/2024

Update Model

Activate

Deactivate

#Log

Update All

16:10:14 02/05/2024<SCADA>: Connecting to server (127.0.0.1:48050)
16:10:14 02/05/2024<SCADA>: Connection status changed to Connected
|

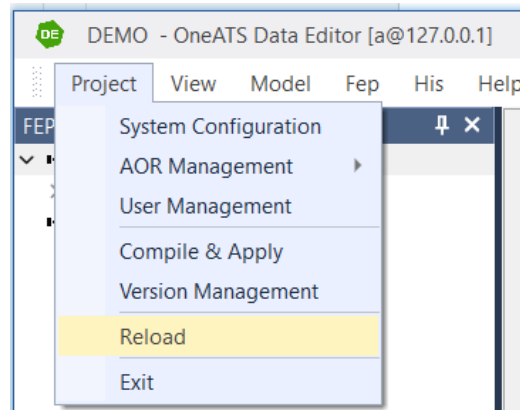
Clear Logs

Close

C.8. Reload project

The Reload project feature allows users to reload the latest model of the project.

Open the **Project** → **Reload** window. Wait for the Reload process to take place.





Contact

ATS Building

74 Ba Trieu Str., Hanoi, Vietnam

T. +84-24-3825 1072

F. +84-24-3825 8037

W. www.ats.com.vn

E. contact@ats.com.vn

